

JAVA PROGRAMMING EXPERIMENT

Basic Information

Experiment No.	Week 07
Date	16/09/26
Title	Method Overriding – Bank (Dynamic Polymorphism)

Objective / Aim

To demonstrate method overriding and dynamic polymorphism in Java by creating a Bank parent class and SBI, HDFC, and ICICI child classes that provide their own implementations of the `getRateOfInterest()` method.

System Requirements

Hardware Requirements

- Computer/Laptop
- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Keyboard and Mouse

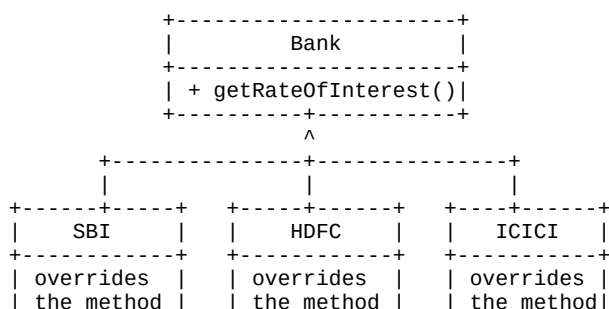
Software Requirements

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK): 17 or 21
- IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Create a parent class named Bank.
2. Define `getRateOfInterest()` in the Bank class.
3. Create SBI, HDFC, and ICICI classes that extend Bank.
4. Override `getRateOfInterest()` in each child class with a different interest rate.
5. In the `main()` method, create Bank references pointing to SBI, HDFC, and ICICI objects.
6. Call `getRateOfInterest()` using each Bank reference.
7. Java selects the overridden method according to the actual object at runtime. This demonstrates dynamic polymorphism.

UML Class Diagram



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Source Code

```
// Parent class
class Bank {

    // Method of parent class
    void getRateOfInterest() {
        System.out.println("Bank Rate of Interest: 5%");
    }
}

// Child class SBI
class SBI extends Bank {

    // Overriding the parent class method
    @Override
    void getRateOfInterest() {
        System.out.println("SBI Rate of Interest: 6.5%");
    }
}

// Child class HDFC
class HDFC extends Bank {

    // Overriding the parent class method
    @Override
    void getRateOfInterest() {
        System.out.println("HDFC Rate of Interest: 7%");
    }
}

// Child class ICICI
class ICICI extends Bank {

    // Overriding the parent class method
    @Override
    void getRateOfInterest() {
        System.out.println("ICICI Rate of Interest: 7.5%");
    }
}

// Main class
public class BankDemo {

    public static void main(String[] args) {

        // Parent class reference referring to SBI object
        Bank b1 = new SBI();

        // Parent class reference referring to HDFC object
        Bank b2 = new HDFC();

        // Parent class reference referring to ICICI object
        Bank b3 = new ICICI();

        // Calling the overridden methods
        b1.getRateOfInterest();
        b2.getRateOfInterest();
        b3.getRateOfInterest();
    }
}
```

Compilation and Execution Commands

```
javac BankDemo.java
java BankDemo
```

Output

```
SBI Rate of Interest: 6.5%
HDFC Rate of Interest: 7%
ICICI Rate of Interest: 7.5%
```

Result

The Java program was successfully compiled and executed. Method overriding and dynamic polymorphism were demonstrated using Bank references referring to SBI, HDFC, and ICICI objects.